# COMP6009 Assignment Part B

## Biogas Plant Intelligent Monitoring and Decision System

**Individual Assessment | 20%**

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## Scenario

A Western Australian red-meat processing facility generates organic residues, wastewater, fats and other biodegradable wastes. These materials are treated using an anaerobic digestion system, where microorganisms break down organic matter to produce biogas. The biogas can subsequently be used for energy recovery, while digestate may be recovered as a useful by-product.

Efficient operation is challenging because waste characteristics and loading conditions vary over time. The true biological condition of the digester cannot be observed directly. Operators instead rely on measurements such as methane production, pH, volatile fatty acids and other indicators, which may be noisy or imperfect.

**Objective:** You are the AI engineer responsible for developing an intelligent monitoring and decision-support system for the facility. Your system must use uncertain sensor information to infer the likely operating condition, track how it changes over time, recommend an appropriate operational response, and evaluate how uncertainty and modelling assumptions affect those recommendations.

## System Model

The true operating condition at time *t* is represented by the hidden state:

$$X_t \in \{\text{Stable, Stressed, Critical}\}$$

The monitoring system receives observations:

$$E_t \in \{\text{Normal, Warning, Alarm}\}$$

The intelligent agent can recommend one of three actions:

$$A_t \in \{\text{Continue, Reduce Feed, Intervene}\}$$

A baseline probability model and sensor datasets are supplied. You must implement the required inference and decision-making algorithms yourself.